

LAPORAN UAS PRAKTIKUM BASIS DATA 2

Dosen Pengampu: Ainurrohman, S.Kom



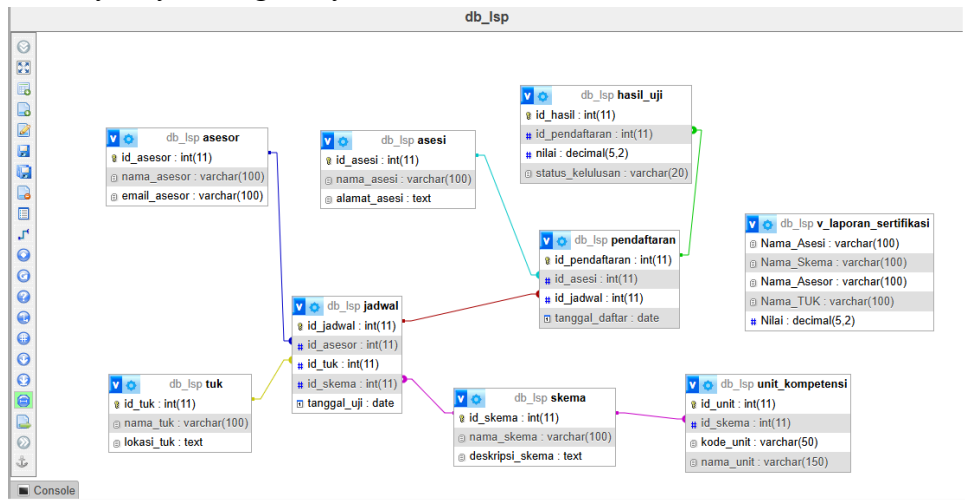
Disusun Oleh:

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POLITEKNIK PURBAYA TEGAL

TEKNIK INFORMATIKA TAHUN 2025/2026

1. Buatlah Entity Relationship Diagram (ERD) minimal terdiri dari tabel berikut.asesor, asesi, skema, unit_kompetensi, tuk, jadwal, pendaftaran, hasil_uji, Lengkapi dengan, Primary Key, Foreign Key.



2. Implementasikan database dengan nama : db_lsp.

```
C:\Command Prompt - mysql -u root
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Win Demo>cd C:\xampp\mysql\bin

C:\xampp\mysql\bin>mysql -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 8
Server version: 10.4.28-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> use mysql;
Database changed
MariaDB [mysql]> CREATE DATABASE db_lsp;
Query OK, 1 row affected (0.002 sec)

MariaDB [mysql]> use db_lsp;
Database changed
```

3. Buat seluruh tabel menggunakan Command-Line Interface (CLI) SQL.

```
Command Prompt - mysql -u root
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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> use mysql;
Database changed
MariaDB [mysql]> CREATE DATABASE db_lsp;
Query OK, 1 row affected (0.002 sec)

MariaDB [mysql]> use db_lsp;
Database changed
MariaDB [db_lsp]> CREATE TABLE asesor (
->   id_asesor INT PRIMARY KEY AUTO_INCREMENT,
->   nama_asesor VARCHAR(100) NOT NULL,
->   email_asesor VARCHAR(100)
-> );
Query OK, 0 rows affected (0.027 sec)

MariaDB [db_lsp]>
MariaDB [db_lsp]> CREATE TABLE asesi (
->   id_asesi INT PRIMARY KEY AUTO_INCREMENT,
->   nama_asesi VARCHAR(100) NOT NULL,
->   alamat_asesi TEXT
-> );
Query OK, 0 rows affected (0.011 sec)

MariaDB [db_lsp]>
MariaDB [db_lsp]> CREATE TABLE skema (
->   id_skema INT PRIMARY KEY AUTO_INCREMENT,
->   nama_skema VARCHAR(100) NOT NULL,
->   deskripsi_skema TEXT
-> );
Query OK, 0 rows affected (0.016 sec)

Command Prompt - mysql -u root
MariaDB [db_lsp]> CREATE TABLE unit_kompetensi (
->   id_unit INT PRIMARY KEY AUTO_INCREMENT,
->   id_skema INT,
->   kode_unit VARCHAR(50),
->   nama_unit VARCHAR(150) NOT NULL,
->   FOREIGN KEY (id_skema) REFERENCES skema(id_skema)
-> );
Query OK, 0 rows affected (0.028 sec)

MariaDB [db_lsp]>
MariaDB [db_lsp]> CREATE TABLE tuk (
->   id_tuk INT PRIMARY KEY AUTO_INCREMENT,
->   nama_tuk VARCHAR(100) NOT NULL,
->   lokasi_tuk TEXT
-> );
Query OK, 0 rows affected (0.011 sec)

MariaDB [db_lsp]>
MariaDB [db_lsp]> CREATE TABLE jadwal (
->   id_jadwal INT PRIMARY KEY AUTO_INCREMENT,
->   id_asesor INT,
->   id_tuk INT,
->   id_skema INT,
->   tanggal_uji DATE,
->   FOREIGN KEY (id_asesor) REFERENCES asesor(id_asesor),
->   FOREIGN KEY (id_tuk) REFERENCES tuk(id_tuk),
->   FOREIGN KEY (id_skema) REFERENCES skema(id_skema)
-> );
Query OK, 0 rows affected (0.033 sec)

MariaDB [db_lsp]>
MariaDB [db_lsp]> CREATE TABLE pendaftaran (
->   id_pendaftaran INT PRIMARY KEY AUTO_INCREMENT,
->   id_asesi INT,
->   id_jadwal INT,
->   tanggal_daftar DATE,
->   FOREIGN KEY (id_asesi) REFERENCES asesi(id_asesi),
->   FOREIGN KEY (id_jadwal) REFERENCES jadwal(id_jadwal)
-> );
Query OK, 0 rows affected (0.038 sec)

MariaDB [db_lsp]>
MariaDB [db_lsp]> CREATE TABLE hasil_uji (
->   id_hasil INT PRIMARY KEY AUTO_INCREMENT,
```

4. Masukkan data minimal 5 asesor, 20 asesi, 5 skema, 20 unit kompetensi, 5 TUK, 10 jadwal, 20 pendaftaran, 20 hasil_uji INSERT INTO

Command Prompt - mysql -u root

```
MariaDB [db_lsp]> INSERT INTO asesor (nama_aseor, email_aseor) VALUES
-> ('Denny Vasanando, M.Kom.', 'denny@lsp.com'),
-> ('Ainurrohman, S.Kom.', 'ainur@lsp.com'),
-> ('Sri Lestari, M.M.', 'lestari@lsp.com'),
-> ('Henny Indriani, M.M.', 'henny@lsp.com'),
-> ('Fahrudin, M.Kom.', 'fahrudin@lsp.com');
Query OK, 5 rows affected (0.081 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]>
MariaDB [db_lsp]> INSERT INTO skema (nama_skema, deskripsi_skema) VALUES
-> ('Junior Web Programmer', 'Skema pemrograman web tingkat dasar'),
-> ('Network Administrator', 'Skema pengelolaan jaringan komputer'),
-> ('Database Designer', 'Skema perancangan basis data'),
-> ('Digital Marketing Specialist', 'Skema pemasaran digital'),
-> ('UI/UX Designer', 'Skema desain antarmuka dan pengalaman pengguna');
Query OK, 5 rows affected (0.006 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]>
MariaDB [db_lsp]> INSERT INTO asesi (nama_asei, alamat_asei) VALUES
-> ('Hanif Nurfajri', 'Semarang'),
-> ('Ananda Firza', 'Kendal'),
-> ('Fakhri Azzaam', 'Demak'),
-> ('Andika Kurniawan', 'Ungaran'),
-> ('M. Ibrahim', 'Salatiga'),
-> ('M. Izani', 'Pekalongan'),
-> ('M. Khirzi', 'Tegal'),
-> ('Fatkhil Dhaifullah', 'Brebek'),
-> ('A. Wahyudin', 'Purwokerto'),
-> ('Asrof', 'Cilacap'),
-> ('M. Salsabil', 'Solo'),
-> ('Gita Savitri', 'Yogyakarta'),
-> ('Hadi Wijaya', 'Magelang'),
-> ('Intan Permata', 'Boyolali'),
-> ('Joko Susilo', 'Sragen'),
-> ('Kartika Sari', 'Klaten'),
-> ('Lukman Hakim', 'Wonosobo'),
-> ('Maya Sari', 'Temanggung'),
-> ('Naufal Alif', 'Pati'),
-> ('Putri Diana', 'Jepara');
Query OK, 20 rows affected (0.002 sec)
Records: 20 Duplicates: 0 Warnings: 0
```

Command Prompt - mysql -u root

```
MariaDB [db_lsp]>
MariaDB [db_lsp]> INSERT INTO unit_kompetensi (id_skema, kode_unit, nama_unit) VALUES
-> (1, 'JWP.01', 'Menerapkan Algoritma Pemrograman'),
-> (1, 'JWP.02', 'Membuat Dokumen Kode Program'),
-> (1, 'JWP.03', 'Menggunakan Struktur Data'),
-> (1, 'JWP.04', 'Membuat Aplikasi Web Dasar'),
-> (2, 'NET.01', 'Merancang Topologi Jaringan'),
-> (2, 'NET.02', 'Mengkonfigurasi Router'),
-> (2, 'NET.03', 'Mengamankan Jaringan Komputer'),
-> (2, 'NET.04', 'Troubleshooting Jaringan'),
-> (3, 'DB.01', 'Mendesain ERD'),
-> (3, 'DB.02', 'Mengimplementasikan Normalisasi'),
-> (3, 'DB.03', 'Menulis Query SQL Lanjutan'),
-> (3, 'DB.04', 'Melakukan Backup dan Recovery'),
-> (4, 'DM.01', 'Membuat Strategi Konten'),
-> (4, 'DM.02', 'Mengelola Media Sosial Bisnis'),
-> (4, 'DM.03', 'Analisis Traffic Web & Ads'),
-> (4, 'DM.04', 'Optimasi SEO On-Page'),
-> (5, 'UI.01', 'Membuat Wireframe'),
-> (5, 'UI.02', 'Melakukan User Research'),
-> (5, 'UI.03', 'Membuat Prototype Interaktif'),
-> (5, 'UI.04', 'Evaluasi Usability Testing');
Query OK, 20 rows affected (0.005 sec)
Records: 20 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]>
MariaDB [db_lsp]> INSERT INTO tuk (nama_tuk, lokasi_tuk) VALUES
-> ('TUK Purbaya 1', 'Ruang A Kampus Politeknik Purbaya'),
-> ('TUK Purbaya 2', 'Lab Komputer Terpadu Purbaya'),
-> ('TUK Mandiri Tegal', 'Jl. Kolonel Sugiono Tegal'),
-> ('TUK Mitra Semarang', 'Jl. Pandanaran Semarang'),
-> ('TUK Mitra Solo', 'Jl. Slamet Riyadi Solo');
Query OK, 5 rows affected (0.002 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```

MariaDB [db_lsp]>
MariaDB [db_lsp]> INSERT INTO jadwal (id_asesor, id_tuk, id_skema, tanggal_uji) VALUES
  -> (1, 1, 1, '2026-08-01'),
  -> (2, 2, 2, '2026-08-02'),
  -> (3, 3, 3, '2026-08-03'),
  -> (4, 4, 4, '2026-08-04'),
  -> (5, 5, 5, '2026-08-05'),
  -> (1, 2, 1, '2026-08-06'),
  -> (2, 3, 2, '2026-08-07'),
  -> (3, 4, 3, '2026-08-08'),
  -> (4, 5, 4, '2026-08-09'),
  -> (5, 1, 5, '2026-08-10');
Query OK, 10 rows affected (0.006 sec)
Records: 10 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]>
MariaDB [db_lsp]> INSERT INTO pendaftaran (id_asesi, id_jadwal, tanggal_daftar) VALUES
  -> (1, 1, '2026-07-25'), (2, 1, '2026-07-25'), (3, 2, '2026-07-25'), (4, 2, '2026-07-25'),
  -> (5, 3, '2026-07-26'), (6, 3, '2026-07-26'), (7, 4, '2026-07-26'), (8, 4, '2026-07-26'),
  -> (9, 5, '2026-07-27'), (10, 5, '2026-07-27'), (11, 6, '2026-07-27'), (12, 6, '2026-07-27'),
  -> (13, 7, '2026-07-28'), (14, 7, '2026-07-28'), (15, 8, '2026-07-28'), (16, 8, '2026-07-28'),
  -> (17, 9, '2026-07-29'), (18, 9, '2026-07-29'), (19, 10, '2026-07-29'), (20, 10, '2026-07-29');
Query OK, 20 rows affected (0.005 sec)
Records: 20 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]>
MariaDB [db_lsp]> INSERT INTO hasil_uji (id_pendaftaran, nilai, status_kelulusan) VALUES
  -> (1, 85.50, 'Kompeten'), (2, 78.00, 'Kompeten'), (3, 90.00, 'Kompeten'), (4, 65.50, 'Belum Kompeten'),
  -> (5, 88.00, 'Kompeten'), (6, 72.50, 'Kompeten'), (7, 95.00, 'Kompeten'), (8, 60.00, 'Belum Kompeten'),
  -> (9, 82.00, 'Kompeten'), (10, 79.50, 'Kompeten'), (11, 91.00, 'Kompeten'), (12, 88.50, 'Kompeten'),
  -> (13, 76.00, 'Kompeten'), (14, 64.00, 'Belum Kompeten'), (15, 89.00, 'Kompeten'), (16, 93.50, 'Kompeten'),
  -> (17, 77.50, 'Kompeten'), (18, 80.00, 'Kompeten'), (19, 84.50, 'Kompeten'), (20, 92.00, 'Kompeten');
Query OK, 20 rows affected (0.012 sec)
Records: 20 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]>

```

5. Buat sebuah VIEW bernama Menampilkan Nama Asesi, Nama Skema, Nama Asesor, Nama TUK

```

MariaDB [db_lsp]> CREATE VIEW v_laporan_sertifikasi AS
  -> SELECT
  ->     ae.nama_asesi AS Nama_Asesi,
  ->     s.nama_skema AS Nama_Skema,
  ->     asr.nama_asesor AS Nama_Asesor,
  ->     t.nama_tuk AS Nama_TUK,
  ->     h.nilai AS Nilai
  -> FROM hasil_uji h
  -> JOIN pendaftaran p ON h.id_pendaftaran = p.id_pendaftaran
  -> JOIN asesi ae ON p.id_asesi = ae.id_asesi
  -> JOIN jadwal j ON p.id_jadwal = j.id_jadwal
  -> JOIN skema s ON j.id_skema = s.id_skema
  -> JOIN asesor asr ON j.id_asesor = asr.id_asesor
  -> JOIN tuk t ON j.id_tuk = t.id_tuk;
Query OK, 0 rows affected (0.018 sec)

MariaDB [db_lsp]>

```

Tampilan perintah lihat view

Command Prompt - mysql -u root

```
MariaDB [db_lsp]> SELECT * FROM v_laporan_sertifikasi;
```

Nama_Asesi	Nama_Skema	Nama_Asesor	Nama_TUK	Nilai
Hanif Nurfajri	Junior Web Programmer	Denny Vasanando, M.Kom.	TUK Purbaya 1	85.50
Hanif Nurfajri	Junior Web Programmer	Denny Vasanando, M.Kom.	TUK Purbaya 1	85.50
Ananda Firza	Junior Web Programmer	Denny Vasanando, M.Kom.	TUK Purbaya 1	78.00
Ananda Firza	Junior Web Programmer	Denny Vasanando, M.Kom.	TUK Purbaya 1	78.00
M. Salsabil	Junior Web Programmer	Denny Vasanando, M.Kom.	TUK Purbaya 2	91.00
M. Salsabil	Junior Web Programmer	Denny Vasanando, M.Kom.	TUK Purbaya 2	91.00
Gita Savitri	Junior Web Programmer	Denny Vasanando, M.Kom.	TUK Purbaya 2	88.50
Gita Savitri	Junior Web Programmer	Denny Vasanando, M.Kom.	TUK Purbaya 2	88.50
Fakhri Azzaam	Network Administrator	Ainurrohman, S.Kom.	TUK Purbaya 2	90.00
Fakhri Azzaam	Network Administrator	Ainurrohman, S.Kom.	TUK Purbaya 2	90.00
Andika Kurniawan	Network Administrator	Ainurrohman, S.Kom.	TUK Purbaya 2	65.50
Andika Kurniawan	Network Administrator	Ainurrohman, S.Kom.	TUK Purbaya 2	65.50
Hadi Wijaya	Network Administrator	Ainurrohman, S.Kom.	TUK Mandiri Tegal	76.00
Hadi Wijaya	Network Administrator	Ainurrohman, S.Kom.	TUK Mandiri Tegal	76.00
Intan Permata	Network Administrator	Ainurrohman, S.Kom.	TUK Mandiri Tegal	64.00
Intan Permata	Network Administrator	Ainurrohman, S.Kom.	TUK Mandiri Tegal	64.00
M. Ibrahim	Database Designer	Sri Lestari, M.M.	TUK Mandiri Tegal	88.00
M. Ibrahim	Database Designer	Sri Lestari, M.M.	TUK Mandiri Tegal	88.00
M. Izani	Database Designer	Sri Lestari, M.M.	TUK Mandiri Tegal	72.50
M. Izani	Database Designer	Sri Lestari, M.M.	TUK Mandiri Tegal	72.50
Joko Susilo	Database Designer	Sri Lestari, M.M.	TUK Mitra Semarang	89.00
Joko Susilo	Database Designer	Sri Lestari, M.M.	TUK Mitra Semarang	89.00
Kartika Sari	Database Designer	Sri Lestari, M.M.	TUK Mitra Semarang	93.50
Kartika Sari	Database Designer	Sri Lestari, M.M.	TUK Mitra Semarang	93.50
M. Khirzi	Digital Marketing Specialist	Henny Indriani, M.M.	TUK Mitra Semarang	95.00
M. Khirzi	Digital Marketing Specialist	Henny Indriani, M.M.	TUK Mitra Semarang	95.00
Fatkhi Dhaifullah	Digital Marketing Specialist	Henny Indriani, M.M.	TUK Mitra Semarang	60.00
Fatkhi Dhaifullah	Digital Marketing Specialist	Henny Indriani, M.M.	TUK Mitra Semarang	60.00
Lukman Hakim	Digital Marketing Specialist	Henny Indriani, M.M.	TUK Mitra Solo	77.50
Lukman Hakim	Digital Marketing Specialist	Henny Indriani, M.M.	TUK Mitra Solo	77.50
Maya Sari	Digital Marketing Specialist	Henny Indriani, M.M.	TUK Mitra Solo	80.00
Maya Sari	Digital Marketing Specialist	Henny Indriani, M.M.	TUK Mitra Solo	80.00
A. Wahyudin	UI/UX Designer	Fahrudin, M.Kom.	TUK Mitra Solo	82.00
A. Wahyudin	UI/UX Designer	Fahrudin, M.Kom.	TUK Mitra Solo	82.00
Asrof	UI/UX Designer	Fahrudin, M.Kom.	TUK Mitra Solo	79.50
Asrof	UI/UX Designer	Fahrudin, M.Kom.	TUK Mitra Solo	79.50
Naufal Alif	UI/UX Designer	Fahrudin, M.Kom.	TUK Purbaya 1	84.50
Naufal Alif	UI/UX Designer	Fahrudin, M.Kom.	TUK Purbaya 1	84.50
Putri Diana	UI/UX Designer	Fahrudin, M.Kom.	TUK Purbaya 1	92.00

6. Gunakan Subquery Menampilkan peserta dengan nilai di atas rata-rata. Menampilkan asesor yang belum memiliki jadwal. Menampilkan skema dengan peserta terbanyak.

- Nilai di atas rata-rata.

```
MariaDB [db_lsp]> SELECT ae.nama_asesi, h.nilai
-> FROM hasil_uji h
-> JOIN pendaftaran p ON h.id_pendaftaran = p.id_pendaftaran
-> JOIN asesori ae ON p.id_asesi = ae.id_asesi
-> WHERE h.nilai > (SELECT AVG(nilai) FROM hasil_uji);
```

nama_asesi	nilai
Hanif Nurfajri	85.50
Fakhri Azzaam	90.00
M. Ibrahim	88.00
M. Khirzi	95.00
A. Wahyudin	82.00
M. Salsabil	91.00
Gita Savitri	88.50
Joko Susilo	89.00
Kartika Sari	93.50
Naufal Alif	84.50
Putri Diana	92.00
Hanif Nurfajri	85.50
Fakhri Azzaam	90.00
M. Ibrahim	88.00
M. Khirzi	95.00
A. Wahyudin	82.00
M. Salsabil	91.00
Gita Savitri	88.50
Joko Susilo	89.00
Kartika Sari	93.50
Naufal Alif	84.50
Putri Diana	92.00

```
22 rows in set (0.018 sec)

MariaDB [db_lsp]>
```

- Menampilkan asesor yang belum memiliki jadwal.

```
MariaDB [db_lsp]> SELECT *
-> FROM asesori
-> WHERE id_asesor NOT IN (SELECT DISTINCT id_asesor FROM jadwal);
Empty set (0.011 sec)
```

- Menampilkan skema dengan peserta terbanyak.

```
MariaDB [db_lsp]> SELECT s.nama_skema, COUNT(p.id_pendaftaran) AS jumlah_peserta
-> FROM skema s
-> JOIN jadwal j ON s.id_skema = j.id_skema
-> JOIN pendaftaran p ON j.id_jadwal = p.id_jadwal
-> GROUP BY s.id_skema, s.nama_skema
-> HAVING COUNT(p.id_pendaftaran) = (
-> SELECT MAX(jml)
-> FROM (
-> SELECT COUNT(p2.id_pendaftaran) AS jml
-> FROM skema s2
-> JOIN jadwal j2 ON s2.id_skema = j2.id_skema
-> JOIN pendaftaran p2 ON j2.id_jadwal = p2.id_jadwal
-> GROUP BY s2.id_skema
-> ) AS sub_max
-> );
```

nama_skema	jumlah_peserta
Junior Web Programmer	8
Network Administrator	8
Database Designer	8
Digital Marketing Specialist	8
UI/UX Designer	8

```
5 rows in set (0.011 sec)

MariaDB [db_lsp]>
```